# Patterns at the workbench

Tungsten Carbide

Name _____ Date _____

Show the equal steps with numbers or small jumps.

**1.** Little counts pens with tiny balls in pairs. What count comes next?

2, 4, 6, ___

Show your thinking.

**2.** A closed drill-bit case has slots grouped in fives. What slot count comes next?

5, 10, 15, ___

Show your thinking.

**3.** Little counts drawn saw teeth in tens. What count comes next?

10, 20, 30, ___

Show your thinking.

# Patterns at the workbench

Tungsten Carbide

Name _____ Date _____

Show the equal steps with numbers or small jumps.

**4.** Little counts pole marks along a snowy path in twos. What count is missing?

12, ___, 16, 18

Show your thinking.

**5.** Little groups scratch-test notes in fives. What count is missing?

20, 25, ___, 35

Show your thinking.

**6.** A workshop notebook counts packed pens in tens. What count is missing?

60, ___, 80, 90

Show your thinking.

WONDER DAILY

# Tungsten Carbide



## A small tip, a big question

A gray tip rested on a cloth. "Could that little piece do a big job?" asked Little Wonderer.





## The ball inside a pen

Little found another clue in a pen. Many ballpoint pens have a tiny tungsten-carbide ball that rolls ink onto paper.





## A line in the soap

Big set soap and a wooden craft stick on a tray. Little guessed what would happen, then watched. Big gently drew the stick across the soap.





## A bend and a break

Does every material bend before it breaks? Big gently bent chalk over the tray. It snapped: chalk is brittle.





## A little grip in the snow

On a snowy walk, Little noticed the tips of their hiking poles. The tip is the small end that meets the ground.





## Hard, tough, or both

Back at the notebook, Big showed a saw-tooth picture. A small carbide tip joined a steel body.



## Make a little book



**1** Fold it into 8 boxes. Open it flat.



**2** Fold short end to short end. Cut the thick line. Open it.



**3** Fold the long way, pictures out. Push the ends in.




**4** Fold into a book. Page 1 goes on the front!

# Give the test its words

Tungsten Carbide · Curiosity day

Name _____ Date _____

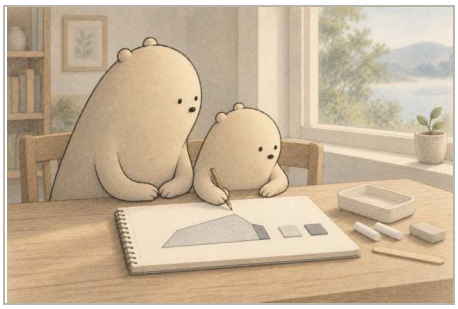
Write one command about the soap test. Say it first if you like.



A line in the soap



A bend and a break



Hard, tough, or both

Draw the scratch Big made in the soap. Add a label.

ask · watch · record · soap · scratch · craft stick · hard · tough

Ask Big to … name one gentle step with the soap and craft stick

Watch the … tell what to notice while Big leads the test

Record the … tell what mark to save in the notebook

Does your sentence tell what to do? Keep one clear command.

# Tungsten Carbide

## 01 What is hiding at the end of a pen?



Many ballpoint pens have a tiny tungsten-carbide ball.

**Look closer.** Look for the tiny round ball where ink meets paper.

**A little explanation.** The tiny ball in many ballpoint pens is tungsten carbide. It rolls ink onto paper and resists wear. People make this material by joining tungsten with carbon, then pressing powder and heating it very hot. This heating step is called sintering. Usually, a little cobalt metal binds the grains together: cemented carbide.

**Try this.** Trace the rolling ball in the picture, then point to the line it leaves.

## 02 What can a scratch tell us?



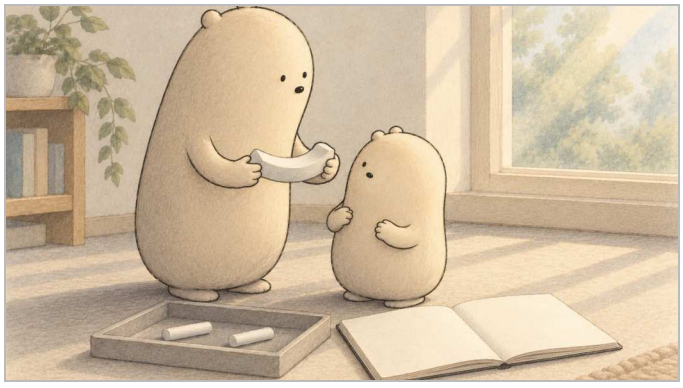
A harder object can leave a scratch in a softer one.

**Look closer.** Big moves the craft stick across soap on a tray. Little watches and records.

**A little explanation.** A harder thing can scratch a softer thing. Hard means resisting scratches and dents. Tungsten carbide is one of the hardest materials people make. It is also heavy for its size. The soap test lets Little see a surface give way.

**Try this.** Predict what mark Big's craft stick will leave in the soap. Record what you notice.

## 03 Does hard always mean tough?



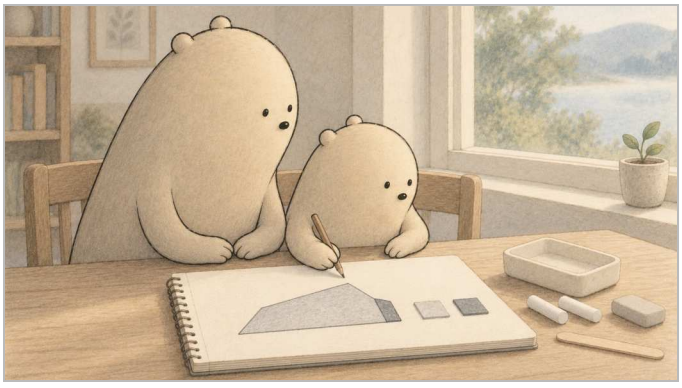
Chalk can snap. A rubber eraser can bend back.

**Look closer.** Compare the chalk pieces with the eraser bending in Big's hands.

**A little explanation.** Tough means resisting cracks or breaks when bumped, bent or dropped. Hard and tough are different. Chalk is brittle: it snaps when bent. An eraser is tough: it bends back. Tungsten carbide is very hard but less tough than steel. It is brittle enough to chip if knocked hard.

**Try this.** Say which test gives a clue about a surface, and which gives a clue about breaking.

## 04 Why give a hard tip a tough body?



Choose a material for the job it needs to do.

**Look closer.** Little's sketch joins a small tip to a larger body.

**A little explanation.** Engineers often join a carbide tip to a tougher steel body, as on saw teeth and some masonry drill bits. The tip resists wear while the steel takes bumps. Carbide also appears in some ski-pole and hiking-pole tips and ice-grip studs. Choose a material for its job: hard, tough, or both.

**Try this.** Choose a job for the tip, then a job for the body.